

DATA SHEET

TDA8443A

I²C-bus controlled YUV/RGB switch

Product specification
Supersedes data of November 1992
File under Integrated Circuits, IC02

1995 Mar 07

Philips Semiconductors



PHILIPS

I²C-bus controlled YUV/RGB switch**TDA8443A****FEATURES**

- Two RGB/YUV selectable clamped inputs with associated synchronization
- RGB/YUV matrix
- 3-state switching with an OFF-state
- Selectable gain
- I²C-bus or non-I²C-bus mode
- Address selection for 7 devices
- Fast switching.

**GENERAL DESCRIPTION**

The TDA8443A is a general purpose two-channel switch for YUV or RGB signals. One channel provides matrixing from RGB to YUV, which can be bypassed.

The IC is controlled via I²C-bus by seven different addresses or can be used in a non-I²C-bus mode. In the non-I²C-bus mode, control of the circuit is achieved by DC voltages.

QUICK REFERENCE DATA

SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
V _P	supply voltage (pin 18)		10.8	12.0	13.2	V
I _P	supply current		—	65	90	mA
RGB/YUV channels						
Z ₁₉₋₂₂	output impedance (pin 19)		—	7	30	Ω
Z ₂₀₋₂₂	output impedance (pin 20)		—	7	30	Ω
Z ₂₁₋₂₂	output impedance (pin 21)		—	7	30	Ω
B	bandwidth	−3 dB; mode 0 or 2	—	25	—	MHz
		+3 dB; mode 0 or 2	—	12	—	MHz
		±3 dB; mode 1	—	10	—	MHz
V _{O(p-p)}	maximum output amplitude of YUV signals (peak-to-peak value)	gain × 1	2.1	—	—	V
		gain × 2	4.2	—	—	V
T _{amb}	operating ambient temperature		0	—	+70	°C

ORDERING INFORMATION

TYPE NUMBER	PACKAGE		
	NAME	DESCRIPTION	VERSION
TDA8443A	DIP24	plastic dual in-line package; 24 leads (600 mil)	SOT101-1

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BLOCK DIAGRAM

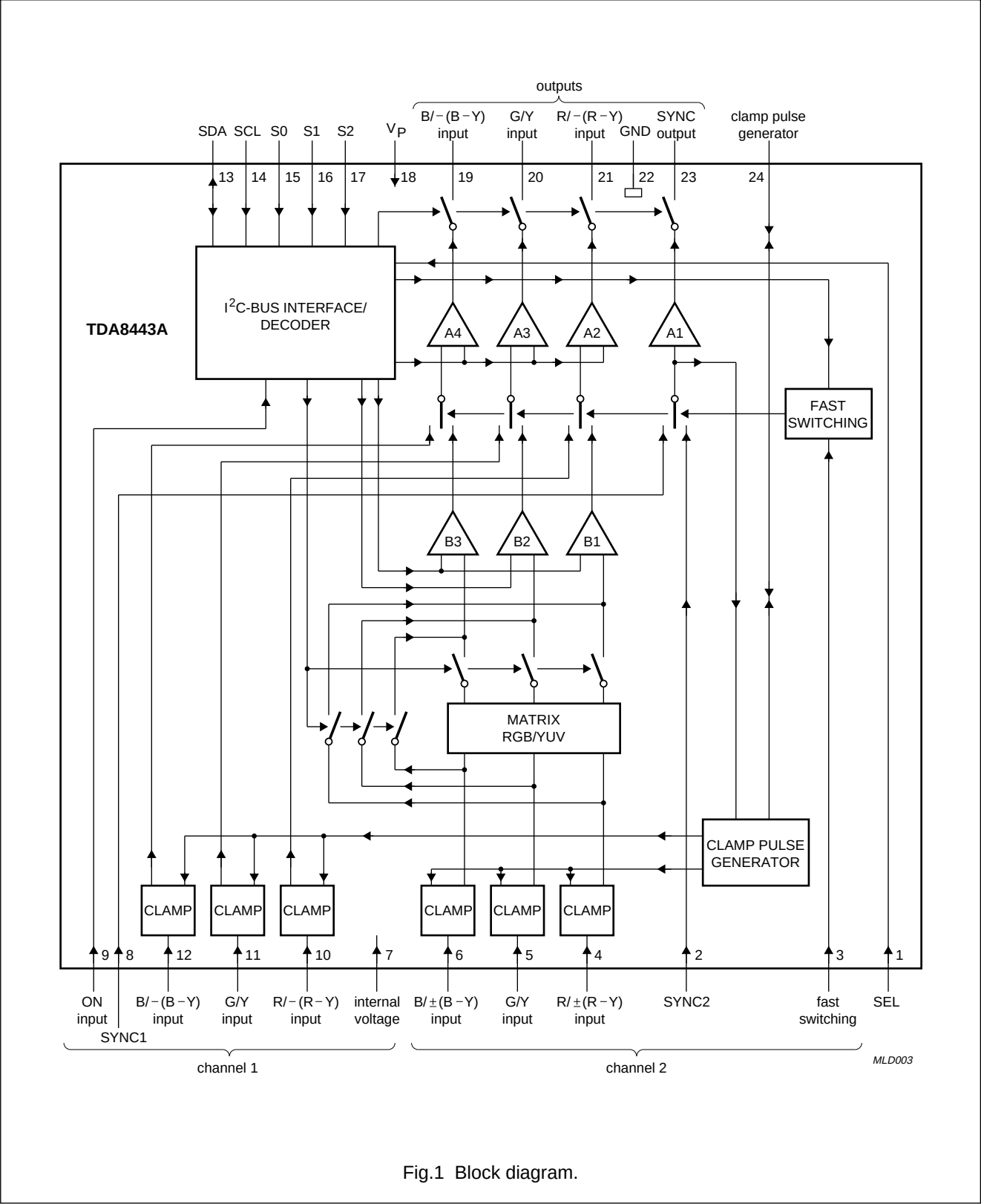


Fig.1 Block diagram.

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PINNING

SYMBOL	PIN	DESCRIPTION
SEL	1	select input (non-I ² C-bus mode only)
SYNC2	2	synchronization input for Channel 2
FS	3	fast switching input
R/±(R-Y)IN	4	R or (R-Y) signal input
G/Y IN	5	G or Y signal input
B/±(B-Y)IN	6	B or (B-Y) signal input
VINT	7	internal voltage supply
SYNC1	8	synchronization input for Channel 1
ON	9	ON input
R/(R-Y)IN	10	R or -(R-Y) signal input
G/Y IN	11	G or Y signal input
B/(B-Y)IN	12	B or -(B-Y) signal input
SDA	13	serial data input/output; I ² C-bus
SCL	14	serial clock input; I ² C-bus
S0	15	address selection input 0
S1	16	address selection input 1
S2	17	address selection input 2
V _P	18	supply voltage
B/(B-Y)OUT	19	B or -(B-Y) signal output
G/Y OUT	20	G or Y signal output
R/(R-Y)OUT	21	R or -(R-Y) signal output
GND	22	ground
SYNC	23	synchronization output
CLAMP	24	clamping pulse generator input/output

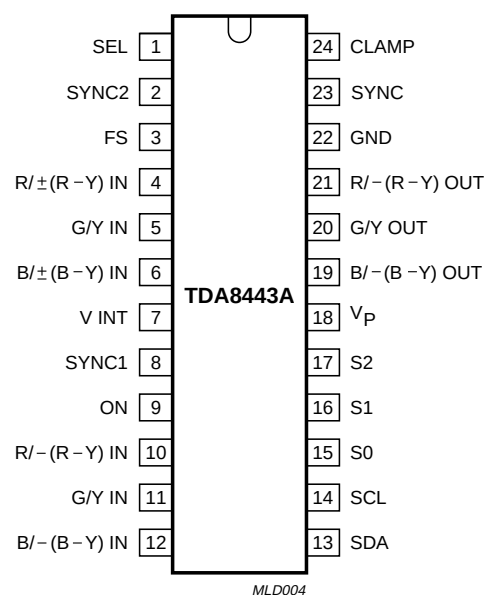


Fig.2 Pin configuration.

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FUNCTIONAL DESCRIPTION

The circuit contains two sets of inputs (see Fig.1). Both channels can receive RGB or YUV signals. Each set of inputs has its own synchronization input, which internally generates a pulse to clamp the inputs. The internal clamping pulse can also be controlled by a signal (e.g. a sandcastle pulse) applied to pin 24. The pulse will occur during the time that the signal at pin 24 is between 5.5 and 6.5 V. If both a sync signal and a pin 24 signal are used the signal should be applied to pin 24 via a 1 k Ω resistor.

RGB signals of Channel 2 can be matrixed to YUV signals.

The outputs can be set in a high impedance OFF state, which allows the use of seven devices in parallel (I²C-bus mode).

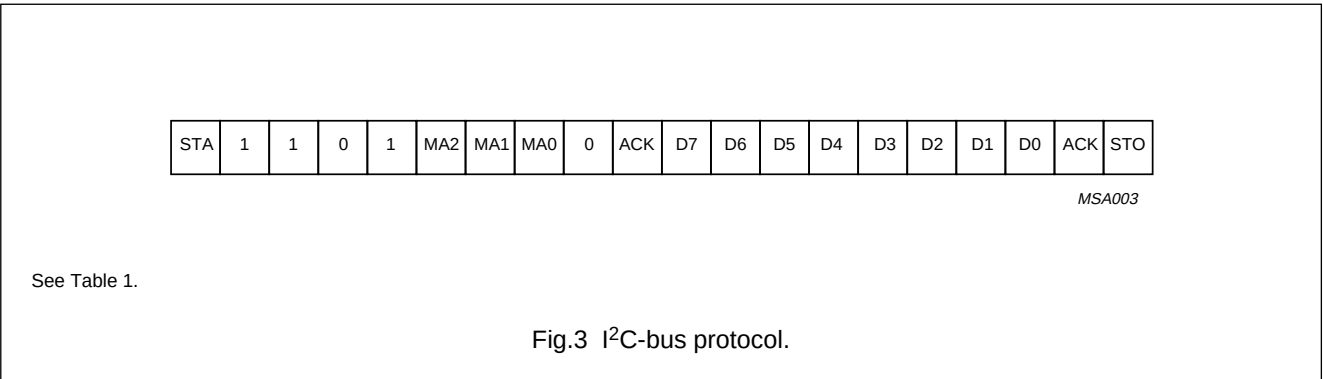
The circuit can be controlled by an I²C-bus compatible microcontroller or directly by DC voltages. The fast switching input can be operated via pin 16 of the peritelevision connector.

I²C-bus mode

The protocol for the devices in I²C-bus mode is shown in Fig.3.

Table 1 Protocol bit description

BIT	DESCRIPTION
STA	start condition
MA2 to MA0	address selection bits; see Table 2
ACK	acknowledge bit
D7	channel selection bit; see Table 3
D6	matrix selection bit; see Table 3
D5 to D3	gain control bits; see Table 4
D2	fast switching priority bit; see Table 5
D1 and D0	output state control bits; see Table 6
STO	stop condition



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Table 2 Address selection

ADDRESS SELECT PINS ⁽¹⁾⁽²⁾			ADDRESS SELECT BITS		
S2 (PIN 17)	S1 (PIN 16)	S0 (PIN 15)	MA2	MA1	MA0
L	L	L	*(3)	*(3)	*(3)
L	L	H	0	0	1
L	H	L	0	1	0
L	H	H	0	1	1
H	L	L	1	0	0
H	L	H	1	0	1
H	H	L	1	1	0
H	H	H	1	1	1

Notes

1. L = LOW level input voltage.
2. H = HIGH level input voltage.
3. * = non-I²C-bus operation.

Table 3 Mode control bits D7 and D6

MODE	D7	D6	FUNCTION
0	0	0	Channel 2 selected, no matrix
1	0	1	Channel 2 selected, matrix active
2	1	0	Channel 1 selected
–	1	1	not allowed

Table 4 Gain setting (see also Table 9)

D5	D4	D3	A1	A2, A3, A4	B1, B3	B2
0	0	0	1	1	–1	0.45
0	0	1	1	1	1	1
0	1	0	not allowed		–	–
0	1	1	1	1	–1	0.45
1	0	0	2	2	–1	0.45
1	0	1	2	1	1	1
1	1	0	2	2	1	1
1	1	1	2	1	–1	0.45

Matrix equations

The relationship between output and input signals of the matrix is as follows:

$$Y = 0.3R + 0.59G + 0.11B$$

$$R - Y = 0.7R - 0.59G - 0.11B$$

$$B - Y = -0.3R - 0.59G + 0.89B$$

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Table 5 Priority/fast switching bit D2

D2	FAST SWITCHING (PIN 3)	MODE
0	X ⁽¹⁾	0 to 2, depending on D7, D6
1	0.4 V	2

Note

1. X = don't care.

Table 6 Output state control bits

D1	D0	PIN 9	FUNCTION
0	X ⁽¹⁾	X ⁽¹⁾	OFF
1	0	L	OFF
1	0	H	ON
1	1	X ⁽¹⁾	ON

Note

1. X = don't care.

Power-on reset

If the circuit is switched on in the I²C-bus mode, all bits of D0 to D7 are set to zero.

Table 7 Non-I²C-bus mode (S2 = S1 = S0 = L)

CONTROL			MODE SWITCHED BY FS (PIN 3)	GAIN SETTINGS		B1, B3	B2
PIN 13	PIN 14	PIN 1		A1	A4, A3, A2		
L	L	L	2 or 0	1	1	1	1
L	L	H	2 or 0	1	2	1	1
L	H	L	2 or 1	1	1	-1	0.45
L	H	H	2 or 0	1	1	-1	0.45
H	L	L	2 or 0	2	1	1	1
H	L	H	2 or 0	2	2	1	1
H	H	L	2 or 1	2	1	-1	0.45
H	H	H	2 or 0	2	1	-1	0.45

Table 8 Fast switching input (pin 3)

FS	MODE SELECTED
≤0.4 V	mode 2
1 to 3 V	mode 0 or mode 1 as set by control

Table 9 ON input (pin 9)

ON	FUNCTION
L	OFF; no output signal; high impedance OFF-state
H	function is determined in Table 7

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LIMITING VALUES

In accordance with the Absolute Maximum System (IEC 134).

SYMBOL	PARAMETER	MIN.	MAX.	UNIT
V _P	supply voltage (pin 18)	–	14	V
V _{I(SDA)}	input voltage (pin 13)	–0.3	14	V
V _{I(SCL)}	input voltage (pin 14)	–0.3	14	V
V _n	input voltage any other pin	–0.3	V _P + 0.3	V
I _{O(max)}	maximum output current	–	20	mA
T _{amb}	operating ambient temperature	0	+70	°C
T _{stg}	IC storage temperature range	–55	+125	°C
T _j	maximum junction temperature	–	+125	°C

CHARACTERISTICS

V_P = 12 V; T_{amb} = 25 °C; unless otherwise specified.

SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
Supply						
V _P	supply voltage (pin 18)		10.8	12.0	13.2	V
I _P	supply current		–	65	90	mA
RGB/YUV channels						
G _{abs}	absolute gain difference (programmed value)		–	0	10	%
G _{rel}	relative gain difference	between Y output and the (R–Y) and (B–Y) channel outputs	–	0	10	%
		between any other two channels	–	0	5	%
I _I	input current		–	0.5	1.0	µA
Z ₁₉₋₂₂	output impedance (pin 19)		–	7	30	Ω
Z ₂₀₋₂₂	output impedance (pin 20)		–	7	30	Ω
Z ₂₁₋₂₂	output impedance (pin 21)		–	7	30	Ω
B	bandwidth	–3 dB; mode 0 or 2	–	25	–	MHz
		+3 dB; mode 0 or 2	–	12	–	MHz
		±3 dB; mode 1	–	10	–	MHz
t _{diff}	mutual time difference at output	all inputs of one source connected together	–	–	25	ns
V _{O(p-p)}	maximum output amplitude of YUV signals (peak-to-peak value)	gain × 1	2.1	–	–	V
		gain × 2	4.2	–	–	V
α _{ct}	crosstalk	note 1; f _i = 5 MHz; between inputs of same source	–	–	–30	dB
		note 1; between same source	–	–	–40	dB
α _{off}	isolation (OFF state)	f _i = 10 MHz	50	–	–	dB

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SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
$G_{\text{diff(p-p)}}$	differential gain at nominal output signals (peak-to-peak value)	$R-Y = 1.05 \text{ V (p-p)}$	–	–	10	%
		$B-Y = 1.33 \text{ V (p-p)}$	–	–	10	%
		$Y = 0.34 \text{ V (p-p)}$	–	–	10	%
S/N	signal-to-noise ratio	nominal input; B = 5 MHz; note 2	50	–	–	dB
SVRR	supply voltage ripple rejection	note 3	30	–	–	dB
V_O	DC output levels during clamping		–	5.3	–	V
Synchronization channels						
G_{diff}	gain difference (programmed value)		–	–	10	%
B	bandwidth	–3 dB	–	50	–	MHz
		+3 dB; gain $\times 1$	–	20	–	MHz
		$\pm 3 \text{ dB; gain } \times 2$	–	13	–	MHz
$V_{i(p-p)}$	input amplitude of sync signal for correct operation of clamp pulse generator (peak-to-peak value)		0.2	–	2.5	V
$ Z_{23-22} $	output impedance (pin 23)		–	20	30	Ω
$V_{o(p-p)}$	maximum undistorted output amplitude (pin 23) (peak-to-peak value)		2.5	–	–	V
V_O	DC output level on top of sync pulse		1.5	1.9	2.4	V
I²C-bus inputs for SDA, SCL						
V_{IH}	HIGH level input voltage		3	–	V_P	V
V_{IL}	LOW level input voltage		–0.3	–	1.5	V
I_{IH}	HIGH level input current		–	–	10	μA
I_{IL}	LOW level input current		–	–	10	μA
I²C-bus output for SDA (open collector)						
V_{OL}	LOW level output voltage	$I_{OL} = 3 \text{ mA}$	–	–	0.4	V
Address selection inputs for S0, S1, S2						
V_{IH}	HIGH level input voltage		3	–	V_P	V
V_{IL}	LOW level input voltage		–0.3	–	0.4	V
I_{IH}	HIGH level input current		–	0	10	μA
I_{IL}	LOW level input current		–50	–10	0	μA
Fast switching input						
V_{IH}	HIGH level input voltage		1	–	3	V
V_{IL}	LOW level input voltage		–0.3	–	0.4	V
I_{IH}	HIGH level input current		–	0	500	μA
I_{IL}	LOW level input current		–100	–	–	μA
t_{sw}	switching time	see Fig.5	–	10	–	ns
t_d	switching delay	see Fig.5	–	20	–	ns

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SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
Select input						
V _{IH}	HIGH level input voltage		3	–	V _P	V
V _{IL}	LOW level input voltage		–0.3	–	0.4	V
I _{IH}	HIGH level input current		–	0	10	μA
I _{IL}	LOW level input current		–50	–10	0	μA
ON input						
V _{IH}	HIGH level input voltage		3	–	V _P	V
V _{IL}	LOW level input voltage		–0.3	–	1.5	V
I _{IH}	HIGH level input current		–	–	10	μA
I _{IL}	LOW level input current		–	–	10	μA

Notes

1. Crosstalk is defined as the unwanted data transfer from an output, driven at nominal level, to other inputs and outputs on the IC and is expressed as a ratio in dBs.
2. Signal-to-noise ratio = $20\log \frac{V_{o(p-p)}}{V_{no(rms)}} (B = 5 \text{ MHz})$
3. Supply voltage ripple rejection = $20\log \frac{V_{RR(supply)}}{V_{RR(at the output)}}$

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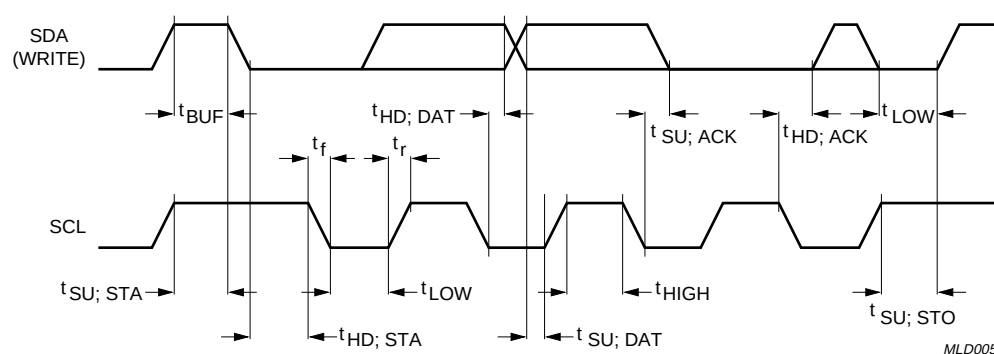
TIMING CHARACTERISTICS

I²C-bus load conditions: 4 k Ω pull-up resistor to +5 V; 200 pF capacitor to GND;
all values are referenced to $V_{IH} = 3$ V and $V_{IL} = 1.5$ V; see Fig.4.

SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.
t_{BUF}	time bus must be free before start		4.7	–	μ s
$t_{SU;STA}$	set-up time for start condition		4.7	–	μ s
$t_{HD;STA}$	hold time for start condition		4.0	–	μ s
t_{LOW}	SCL and SDA LOW time		4.7	–	μ s
t_{HIGH}	SCL HIGH time		4.0	–	μ s
t_r	SCL and SDA rise time		–	1.0	μ s
t_f	SCL and SDA fall time		–	0.3	μ s
$t_{SU;DAT}$	data set-up time (write)		250	–	ns
$t_{HD;DAT}$	data hold time (write)	note 1	1.0	–	μ s
$t_{SU;ACK}$	acknowledge set-up time		–	2	μ s
$t_{HD;ACK}$	acknowledge hold time		0	–	μ s
$t_{SU;STO}$	set-up time for stop condition		4.7	–	μ s

Note

1. Timing $t_{HD;DAT}$ deviates from the I²C-bus specification. After reset has been activated, a delay of 50 μ s must occur before transmission may be resumed.

Fig.4 I²C-bus timing diagram.

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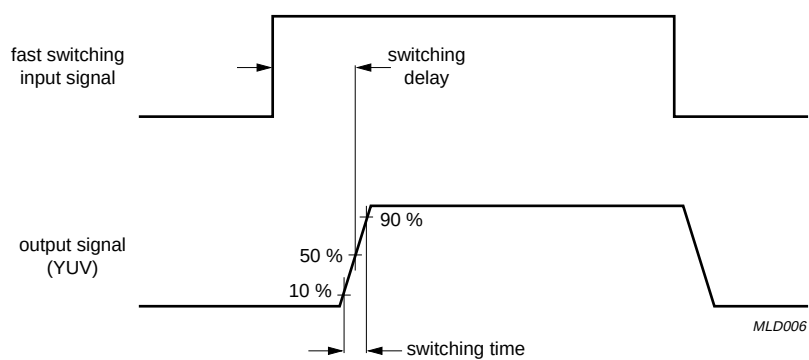


Fig.5 Fast switching signal diagram.

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APPLICATION INFORMATION

Table 10 Channel input/output information

INPUT 1	INPUT 2	OUTPUT	MODE	D5	D4	D3
Y = 0.34 V U = -1.33 V V = -1.05 V S = 0.3 V	–	Y = 0.34 V U = -1.33 V V = -1.05 V S = 0.6 V	2	1	1	1
–	R = 0.75 V G = 0.75 V B = 0.75 V S = 0.3 V	Y = 0.34 V U = -1.33 V V = -1.05 V S = 0.6 V	1	1	1	1
Y = 0.34 V U = -1.33 V V = -1.05 V S = 0.3 V	–	Y = 0.68 V U = -2.66 V V = -2.10 V S = 0.6 V	2	1	0	0
–	R = 0.75 V G = 0.75 V B = 0.75 V S = 0.3 V	Y = 0.68 V U = -2.66 V V = -2.10 V S = 0.6 V	1	1	0	0
Y = 0.34 V U = -1.33 V V = -1.05 V S = 0.3 V	–	Y = 0.34 V U = -1.33 V V = -1.05 V S = 0.6 V	2	1	0	1
–	Y = 0.34 V U = -1.33 V V = -1.05 V S = 0.3 V	Y = 0.34 V U = -1.33 V V = -1.05 V S = 0.6 V	0	1	0	1
Y = 0.34 V U = -1.33 V V = -1.05 V S = 0.3 V	–	Y = 0.68 V U = -2.66 V V = -2.10 V S = 0.6 V	2	1	1	0
–	Y = 0.34 V U = -1.33 V V = -1.05 V S = 0.3 V	Y = 0.68 V U = -2.66 V V = -2.10 V S = 0.6 V	0	1	1	0

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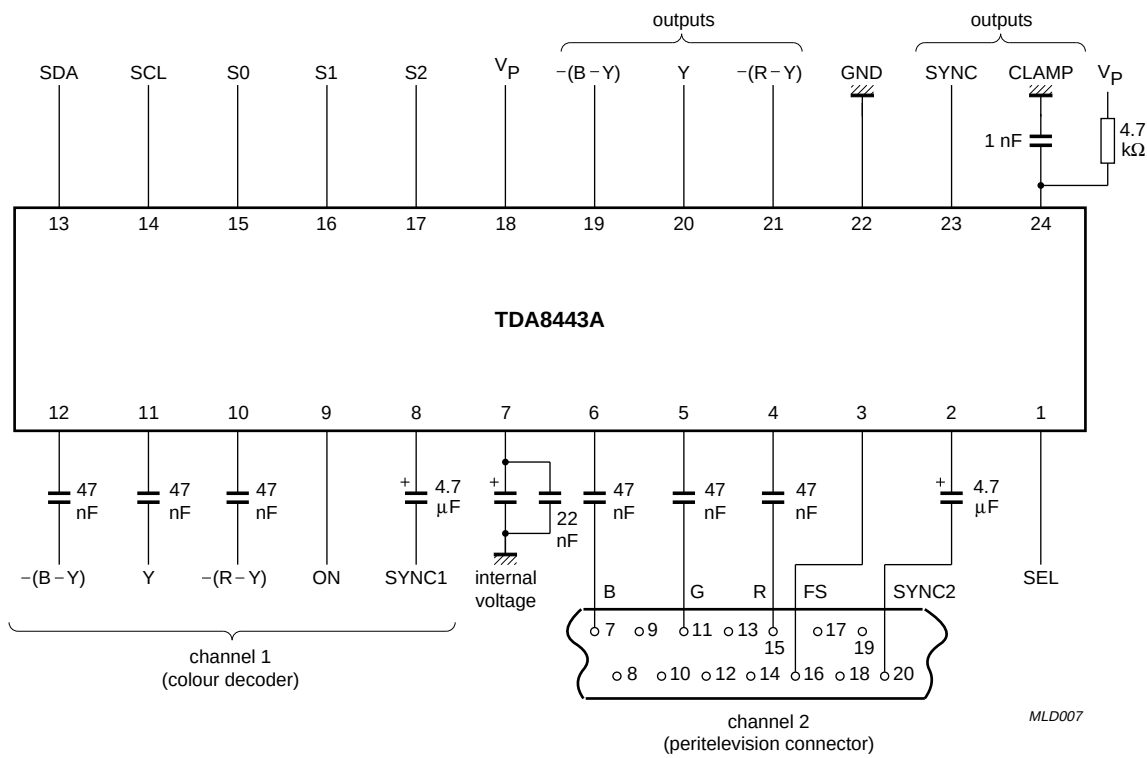


Fig.6 Application diagram (example).

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Input clamps

The R, G, B respectively (R–Y), Y and (B–Y) video signals are AC-coupled to the IC where they are clamped on the black level. The timing information for this clamping action is derived from the associated synchronization signal SYNC, which could also consist of the composite video information signal CVBS. The syncsignal is AC-coupled to the IC where it is clamped on top-sync level, information obtained from this action is used to generate the clamp pulses.

The clamp pulses can be generated in two ways:

1. Using the sync information (internal clamping)

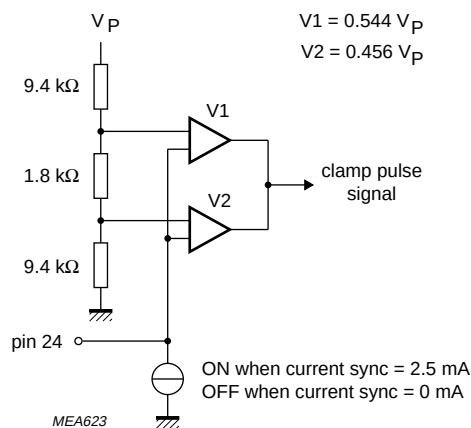
The sync information is clamped on top-sync and the information obtained from this action is used to switch an internal current source at pin 24.

Pin 24 should be connected to V_P via a 4.7 k Ω resistor, and a 1 nF capacitor to ground. During video scan the voltage at pin 24 will be HIGH (equals positive supply voltage). During the synchronization pulses the voltage at pin 24 will drop to zero because of the current sink (2.5 mA).

When the synchronization pulse is over, the current source is switched off and the voltage at pin 24 will rise to its higher level. Because of the time constant at pin 24, the restoration will take some microseconds. The voltage at pin 24 is also sensed internally and at the time it is between $0.456V_P$ and $0.544V_P$, a time pulse is generated and used for the clamping action.

2. Using a sandcastle pulse (external clamping)

If an associated sandcastle pulse is available, it can also be used as a clamping pulse. In this event the sandcastle pulse should be connected to pin 24, the top of the clamping pulse should be between $0.544V_P$ and $0.456V_P$. The timing of the internal clamping pulse will be equal to the timing of the higher part of the sandcastle pulse. If the sync signal is also connected, the current sink will also become active during the synchronization pulses. This means that the sandcastle pulse should be connected to pin 24 via a 1 k Ω dropping resistor. In this event only the sandcastle pulse at pin 24 will be influenced during sync pulses, but the sandcastle pulse at the sandcastle source will be unchanged.



Tolerance on V1 and V2 is given by $\Delta R/R$ and $\Delta V_P/V_P$. The diffusion process gives $\Delta R/R$ (max) = 1.5%.

Fig.7 Clamping circuit.

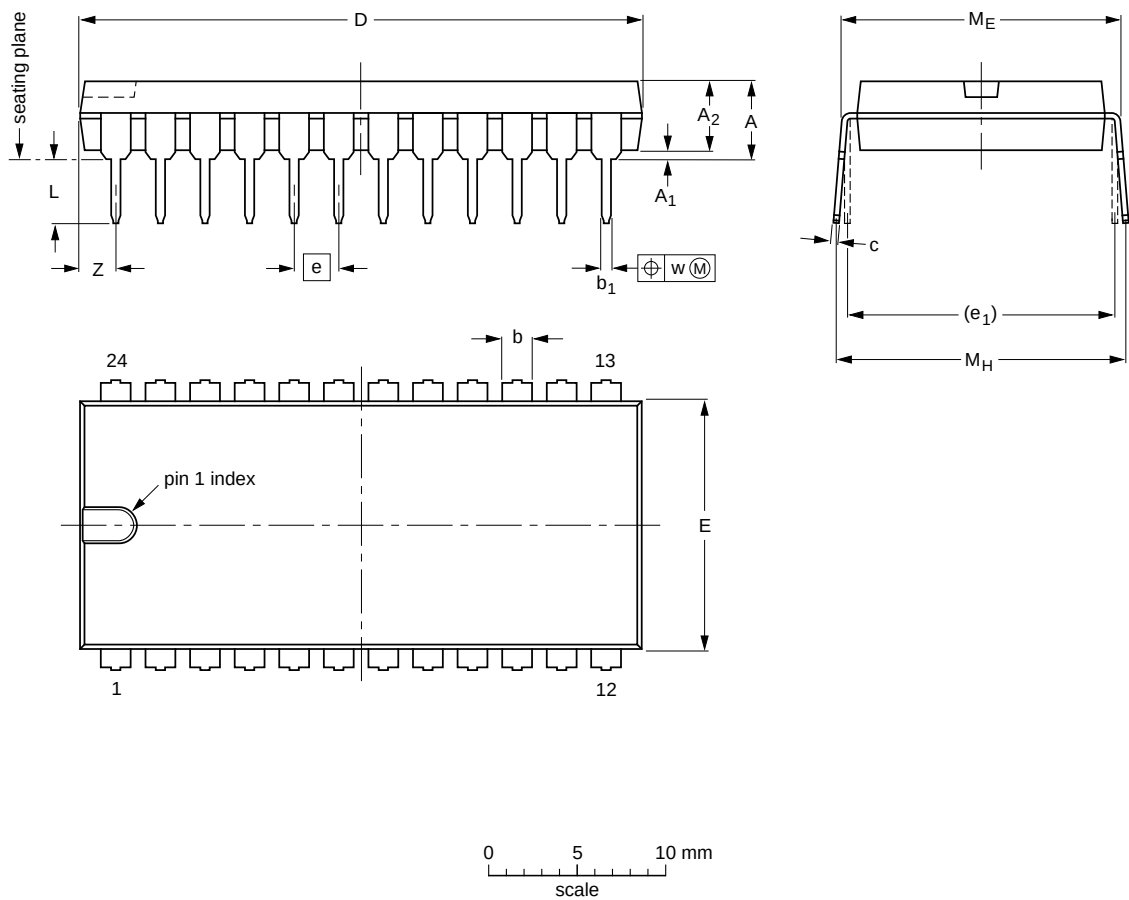
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PACKAGE OUTLINE

DIP24: plastic dual in-line package; 24 leads (600 mil)

SOT101-1

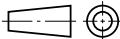


DIMENSIONS (inch dimensions are derived from the original mm dimensions)

UNIT	A max.	A ₁ min.	A ₂ max.	b	b ₁	c	D ⁽¹⁾	E ⁽¹⁾	e	e ₁	L	M _E	M _H	w	Z ⁽¹⁾ max.
mm	5.1	0.51	4.0	1.7 1.3	0.53 0.38	0.32 0.23	32.0 31.4	14.1 13.7	2.54	15.24	3.9 3.4	15.80 15.24	17.15 15.90	0.25	2.2
inches	0.20	0.020	0.16	0.066 0.051	0.021 0.015	0.013 0.009	1.26 1.24	0.56 0.54	0.10	0.60	0.15 0.13	0.62 0.60	0.68 0.63	0.01	0.087

Note

1. Plastic or metal protrusions of 0.25 mm maximum per side are not included.

OUTLINE VERSION	REFERENCES				EUROPEAN PROJECTION	ISSUE DATE
	IEC	JEDEC	EIAJ			
SOT101-1	051G02	MO-015AD				92-11-17 95-01-23

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SOLDERING**Plastic dual in-line packages**

BY DIP OR WAVE

The maximum permissible temperature of the solder is 260 °C; this temperature must not be in contact with the joint for more than 5 s. The total contact time of successive solder waves must not exceed 5 s.

The device may be mounted up to the seating plane, but the temperature of the plastic body must not exceed the specified storage maximum. If the printed-circuit board has been pre-heated, forced cooling may be necessary immediately after soldering to keep the temperature within the permissible limit.

REPAIRING SOLDERED JOINTS

Apply a low voltage soldering iron below the seating plane (or not more than 2 mm above it). If its temperature is below 300 °C, it must not be in contact for more than 10 s; if between 300 and 400 °C, for not more than 5 s.

DEFINITIONS

Data sheet status	
Objective specification	This data sheet contains target or goal specifications for product development.
Preliminary specification	This data sheet contains preliminary data; supplementary data may be published later.
Product specification	This data sheet contains final product specifications.
Limiting values	
Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.	
Application information	
Where application information is given, it is advisory and does not form part of the specification.	

LIFE SUPPORT APPLICATIONS

These products are not designed for use in life support appliances, devices, or systems where malfunction of these products can reasonably be expected to result in personal injury. Philips customers using or selling these products for use in such applications do so at their own risk and agree to fully indemnify Philips for any damages resulting from such improper use or sale.

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NOTES

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NOTES

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